

## Project Planning Phase

| Sprint   | Duration | Start Date | End Date   | Goals / Deliverables                                       | Estimation | Notes                                    |
|----------|----------|------------|------------|------------------------------------------------------------|------------|------------------------------------------|
| Sprint 1 | 2        | 2026-06-10 | 2026-06-23 | Requirement analysis, environment setup, data collection   | 10         | Setup network data capture tools         |
| Sprint 2 | 2        | 2026-06-24 | 2026-07-07 | Data preprocessing pipeline, feature engineering           | 12         | Develop scripts for feature extraction   |
| Sprint 3 | 3        | 2026-07-08 | 2026-07-28 | Initial AI model development (classical ML baseline)       | 15         | Train/test Random Forest, SVM            |
| Sprint 4 | 3        | 2026-07-29 | 2026-08-18 | Deep learning model design and prototyping                 | 18         | LSTM/CNN models for anomaly detection    |
| Sprint 5 | 2        | 2026-08-19 | 2026-09-01 | Integration of detection engine with real-time data stream | 14         | Connect models with data pipeline        |
| Sprint 6 | 2        | 2026-09-02 | 2026-09-15 | Alerting system and dashboard UI prototype                 | 12         | Setup notification and visualization     |
| Sprint 7 | 2        | 2026-09-16 | 2026-09-29 | System testing, performance tuning, and documentation      | 12         | Test for false positives, optimize speed |
| Sprint 8 | 1        | 2026-09-30 | 2026-10-06 | Final deployment and user training                         | 8          | Deploy system and train users            |